

California Polytechnic State University, San Luis Obispo

Project Proposal

CT-Based Tumor Segmentation
with Convolutional Neural Networks

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For our final project, we propose to develop a deep learning system for medical image segmentation of ultrasound breast cancer images.

Our primary objective is to identify and localize tumor regions using image-level class labels, with the potential to extend the approach to pixel-level segmentation if feasible. This task is particularly meaningful because tumors exhibit significant variability in size, shape, and contrast. Effective segmentation models must therefore address challenges such as class imbalance in terms of a large amount of background versus foreground pixels. and boundary ambiguity.

We are interested in this project because accurate tumor segmentation plays a critical role in treatment planning and quantitative assessment within clinical workflows. At the same time, it presents substantial technical challenges due to noisy imaging data, inter-patient variability, and limited availability of annotated datasets.

Our approach will involve designing and implementing an end-to-end convolutional neural network (CNN)–based segmentation pipeline. We will evaluate its performance using appropriate metrics and iteratively refine the model through at least one improvement cycle based on experimental results.

Background

Deep learning has become a dominant approach for medical segmentation, especially with U-Net style architecture for dense pixel level prediction. After the success of the U-Net architecture, there have been further improvements with residual blocks [Zhang et al.] and attention gates [Oktay et al.] being implemented into the original U-Net architecture. Recent benchmarks studies

also show that training strategies such as augmentation and different loss functions are as important as architecture choice.

Data and methods

We plan to use ultrasound-based dataset from our candidate list, with final selection after some discussion and feedback. Here is a link to the proposed dataset:

<https://www.kaggle.com/datasets/aryashah2k/breast-ultrasound-images-dataset>

This dataset consists of benign and malignant tumours in ultrasound images of several different patients. Moreover, there are ultrasounds of breasts without any tumors. This will allow our model to also be robust to detect if there are any tumors at all.

So far our plan is to

- Preprocess ultrasound images (e.g. augmentation, normalization, resizing/cropping)
- Train a baseline U-Net CNN for segmentation
- Build an improved model (e.g. Residual blocks or attention gate variant)
- Compare baseline and improved versions under the same split and training data

Evaluation

We will evaluate segmentation quality using:

- Precision/Recall (or sensitivity/specificity) for tumor pixels
- Intersection over Union (IoU) and dice coefficient

We will compare:

- Baseline U-Net vs Improved model
- Different loss functions
- Effects of augmentation

Expected Outcomes

By the end of the project, we expect:

- A working breast cancer ultrasound segmentation pipeline with a quantitative baseline and a revised model with measurable improvement
- Better tumor boundary localization in qualitative overlays
- If time allows, clear analysis of failure cases (small tumors, low contrast, false positives)

Ethical Implications

This project has important ethical considerations:

- Clinical risk: model errors could mislead decisions, research only not for diagnosis
- Bias/generalization: dataset composition may not represent all populations/scanners
- Data privacy: we will use public de-identified data and follow dataset licenses
- Responsible claims: we will report limitations and avoid overstating clinical readiness

References

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